

Data Quality Strategy

Snowflake Automated QA Framework

Executive Vision

Trusted data is the foundation of modern AI and analytics. Our strategy transitions from reactive manual checks to a metadata-driven, automated validation framework embedded directly into the Snowflake data lifecycle.

Scalability

Metadata-driven testing logic.

Reliability

Quality gates in CI/CD.

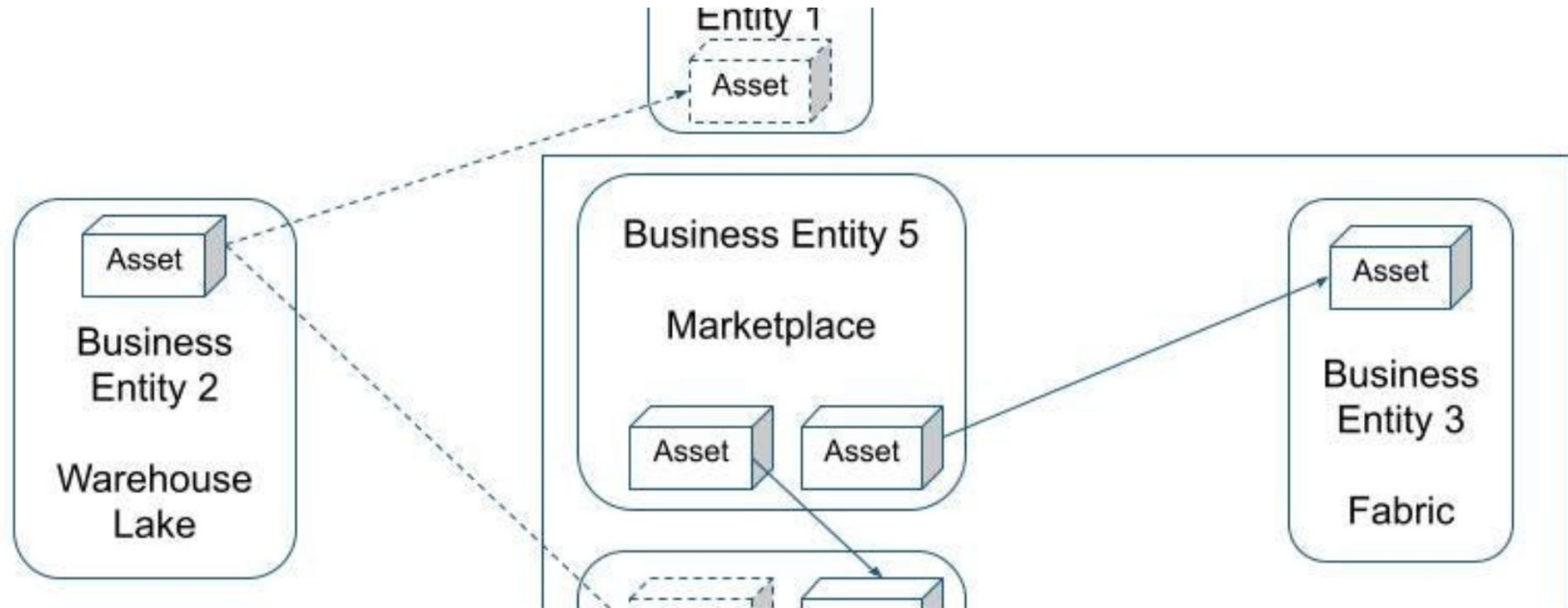
Observability

Continuous monitoring loops.

Core Quality Dimensions

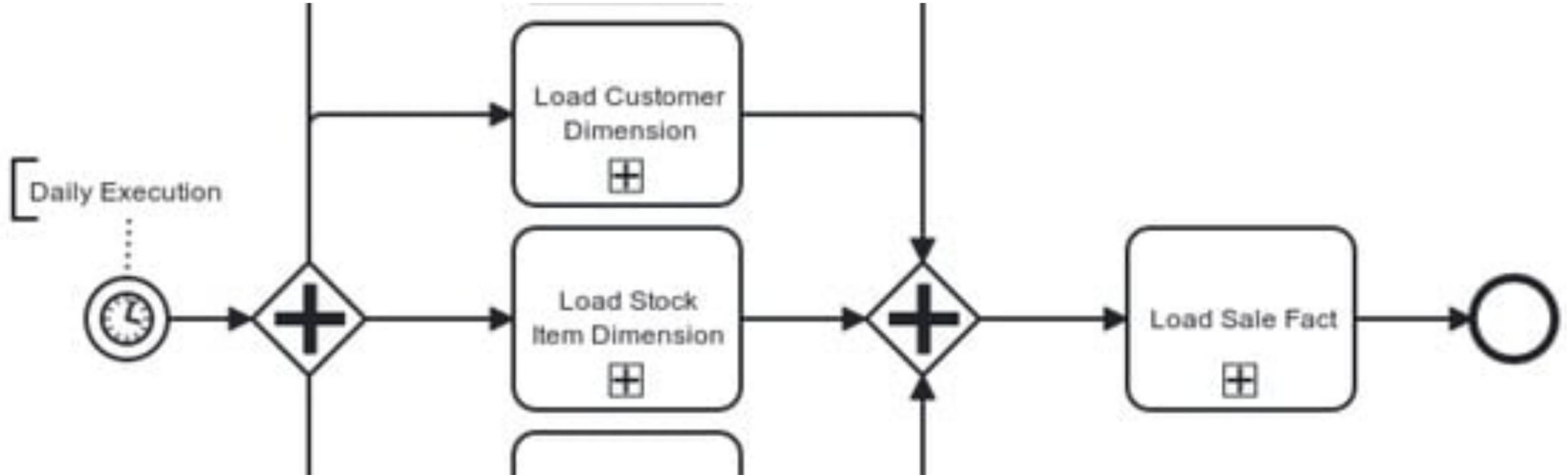
- **Accuracy:** Data reflects reality.
- **Completeness:** No critical gaps.
- **Consistency:** Uniformity across systems.
- **Validity:** Adherence to schemas.
- **Uniqueness:** No duplicates.
- **Timeliness:** Data available on time.

Framework Architecture



A tiered approach: Data is validated at the ingestion (Raw), transformation (Silver), and consumption (Gold) layers using Snowflake's native Data Metric Functions (DMFs).

Automated QA Workflow



Automation triggers validation scripts during every pipeline execution. Failures are captured in a centralized audit table for automated alerting.

Testing Layers

Unit Testing

Data validation at ingestion (e.g., null checks, schema enforcement).

Integration Testing

Verification across joins and transformation logic.

Metadata-Driven Tests

Instead of hardcoded scripts, we use a central Registry Table to define validation rules.

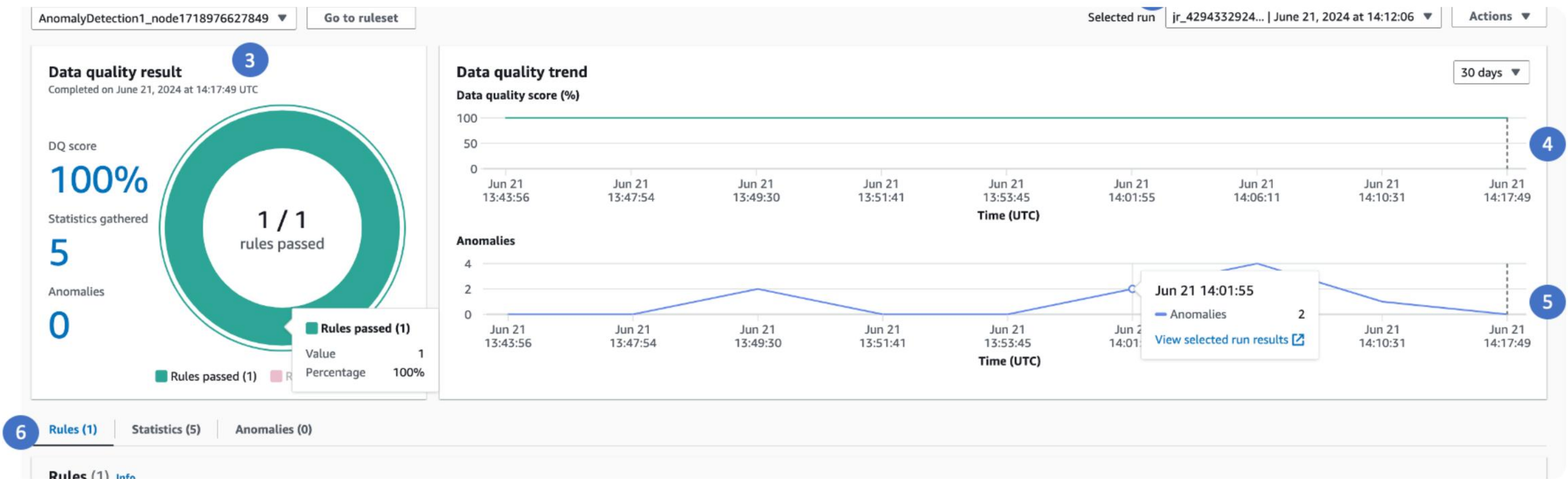
Registry Entry Example: {Rule: "NotNull", Column: "ORDER_ID", Table: "SALES_FACT"}

Calculating Data Trust

$$DQ = \frac{\text{Passed Rules}}{\text{Total Rules}} \times 100$$

This score is visualized in real-time dashboards for stakeholders.

Visibility & Scorecards



Efficiency Gains

Manual QA

High latency, error-prone, limited coverage, high cost.

Automated Framework

Real-time, scalable, exhaustive coverage, proactive.

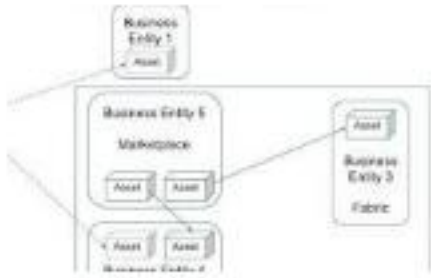
Next Steps



Questions & Discussion

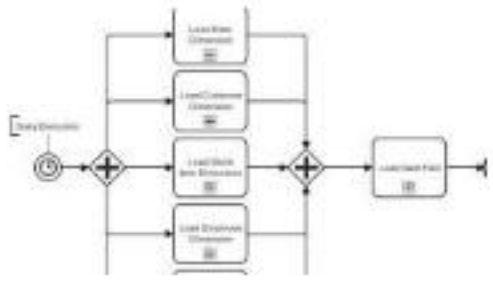
Thank you for your time. We look forward to building a resilient data foundation together.

Image Sources



https://miro.medium.com/v2/resize:fit:1400/1*vJXH0pZ9GN9ONI99ZZX-mg.jpeg

Source: medium.com



https://www.mdpi.com/data/data-10-00097/article_deploy/html/images/data-10-00097-g001-550.jpg

Source: www.mdpi.com



<https://docs.aws.amazon.com/images/glue/latest/dg/images/data-quality-visualize-data-quality-metrics.png>

Source: docs.aws.amazon.com



https://static.vecteezy.com/system/resources/previews/055/143/521/non_2x/abstract-network-of-interconnected-nodes-representing-communication-technology-and-data-connections-in-a-digital-web-png.png

Source: www.vecteezy.com